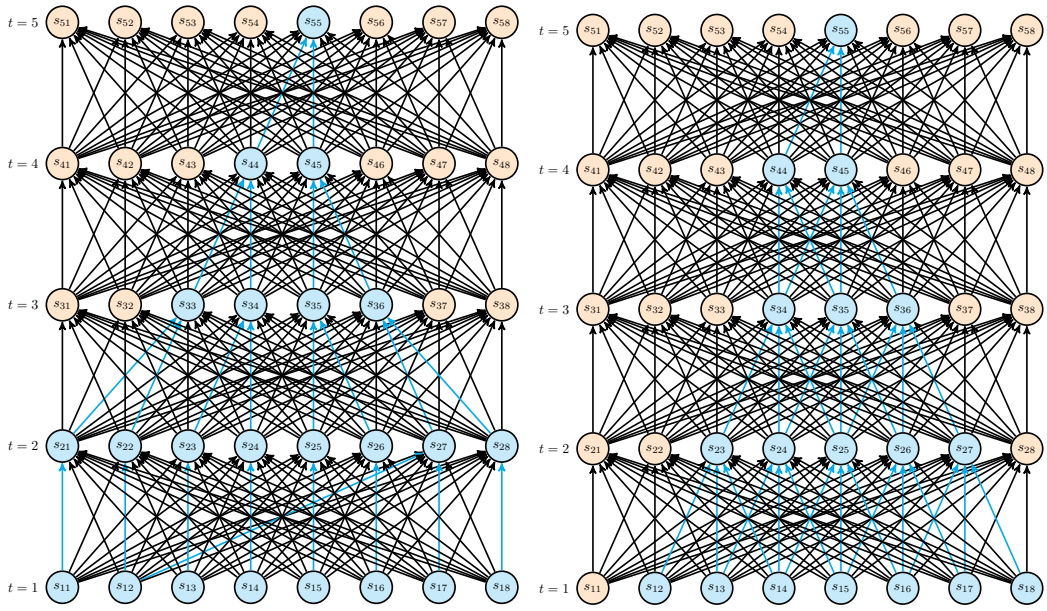

Figure A1: Examples of how affected states (cyan) affect states in a previous time step (indicated by cyan edges) starting with $n_5 = 1$ (one affected state). The left panel shows no overlap since affected states in s_{t-1} connect only to one affected state in s_t. The right panel shows some overlap since affected states in s_{t-1} connect to multiple affected states in s_t.

Proof. The "backward induction algorithm" [95, 96] gives with $\boldsymbol{V}_{T+1}^{\pi} = \boldsymbol{0}$ and on-site variance $\boldsymbol{\omega}_{T+1} = \boldsymbol{0}$:

$$\boldsymbol{V}_t^{\pi} \;=\; \sum_{k=t}^{T} \prod_{\tau=t}^{k-1} \boldsymbol{P}_\tau \, \boldsymbol{\omega}_k \;, \tag{A238}$$

where we define $\prod_{\tau=t}^{t-1} \boldsymbol{P}_\tau = \boldsymbol{I}$ and $[\boldsymbol{\omega}_k]_{(s_k,a_k)} = \omega(s_k, a_k)$.
Since the product of two row-stochastic matrices is a row-stochastic matrix according to Lemma A4, $\boldsymbol{P}_{t \leftarrow k} = \prod_{\tau=t}^{k-1} \boldsymbol{P}_\tau$ is a row-stochastic matrix. Since $\|\boldsymbol{P}_{t \leftarrow k}\|_\infty = 1$ according to Lemma A5, each on-site variance $\boldsymbol{\omega}_k$ with $t \leqslant k$ can have large effects on $\boldsymbol{V}_t^{\pi}$. Using the row-stochastic matrices $\boldsymbol{P}_{t \leftarrow k}$, we can reformulate the variance:

$$\boldsymbol{V}_t^{\pi} \;=\; \sum_{k=t}^{T} \boldsymbol{P}_{t \leftarrow k} \, \boldsymbol{\omega}_k \;, \tag{A239}$$

with $\|\boldsymbol{P}_{t \leftarrow k}\|_\infty = 1$. The on-site variance $\boldsymbol{\omega}_k$ at step k increases all variances $\boldsymbol{V}_t^{\pi}$ with $t \leqslant k$.
Next we proof the second part of the theorem, which considers the growth of a_k. To compute a_k we first have to know n_t. For computing n_{t-1} from n_t, we want to know how many states are affected in s_{t-1} if n_t states are affected in s_t. The answer to this question is the expected coverage when searching a document collection using a set of independent computers [19]. We follow the approach of Cox et al. [19]. The minimal number of affected states in s_{t-1} is c_t, where each of the c_t affected states in s_{t-1} connects to each of the n_t states in s_t (maximal overlap). The maximal number of affected states in s_{t-1} is $c_t n_t$, where each affected state in s_{t-1} connects to only one affected state in s_t (no overlap). We consider a single state in s_t. The probability of a state in s_{t-1} being connected to this single state in s_t is c_t / N_{t-1} and being not connected to this state in s_t is $1 - c_t / N_{t-1}$. The probability of a state in s_{t-1} being not connected to any of the n_t affected states in s_t is

$$\left(1 \;-\; \frac{c_t}{N_{t-1}}\right)^{n_t} \;. \tag{A240}$$

The probability of a state in s_{t-1} being at least connected to one of the n_t affected states in s_t is

$$1 \;-\; \left(1 \;-\; \frac{c_t}{N_{t-1}}\right)^{n_t} \;. \tag{A241}$$

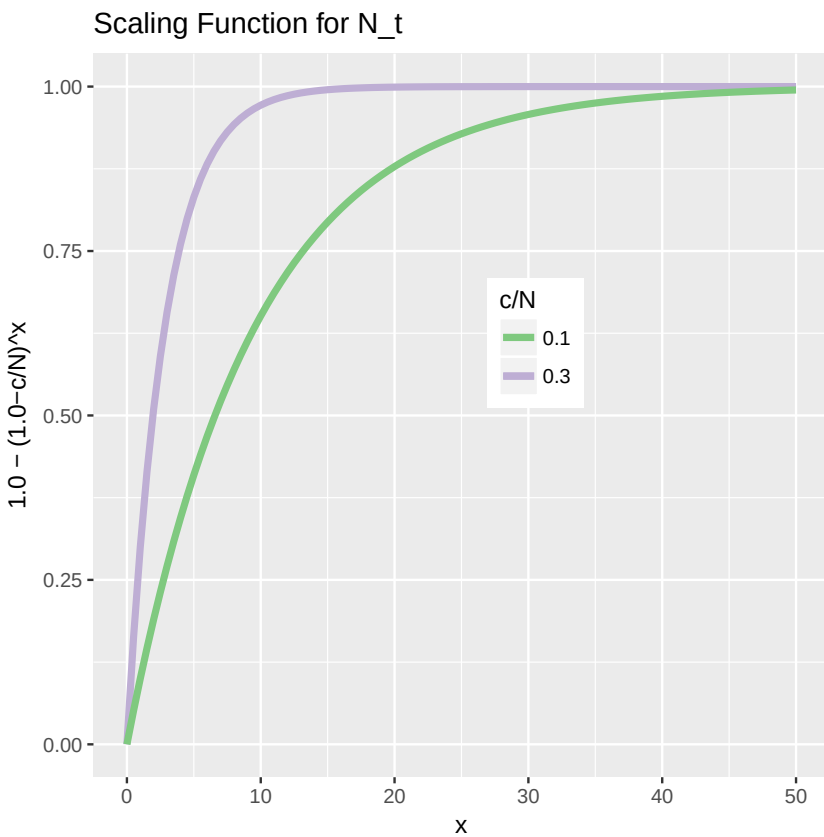

Figure A2: The function $\left(1 - \left(1 - \frac{c_t}{N_{t-1}}\right)^{n_t}\right)$ which scales N_{t-1} in Theorem A10. This function determines the growth of a_k, which is exponentially at the beginning, and then linearly when the function approaches 1.

Thus, the expected number of distinct states in s_{t-1} being connected to one of the n_t affected states in s_t is

$$n_{t-1} \;=\; \left(1 \;-\; \left(1 \;-\; \frac{c_t}{N_{t-1}}\right)^{n_t}\right) \; N_{t-1} \; . \tag{A242}$$

The number a_k of affected states by $\boldsymbol{\omega}_k$ is

$$a_k \;=\; \sum_{t=0}^{k} \left(1 \;-\; \left(1 \;-\; \frac{c_t}{N_{t-1}}\right)^{n_t}\right) \; N_{t-1} \; . \tag{A243}$$

□

Corollary A2. *For small k, the number a_k of states affected by the on-site variance $\boldsymbol{\omega}_k$ at step k grows exponentially with k by a factor of $\bar{c}$:*

$$a_k \;>\; \bar{c}^k \; . \tag{A244}$$

For large k and after some time $t > \hat{t}$, the number a_k of states affected by $\boldsymbol{\omega}_k$ grows linearly with k with a factor of $\bar{N}$:

$$a_k \;\approx\; a_{\hat{t}-1} \;+\; (k - \hat{t} + 1) \; \bar{N} \; . \tag{A245}$$

Proof. For small n_t with $\frac{c_t n_t}{N_{t-1}} \ll 1$, we have

$$\left(1 \;-\; \frac{c_t}{N_{t-1}}\right)^{n_t} \;\approx\; 1 \;-\; \frac{c_t \; n_t}{N_{t-1}} \; , \tag{A246}$$

thus

$$n_{t-1} \;\approx\; c_t \, n_t \; . \tag{A247}$$

For large N_{t-1} compared to the number of connections c_t of a single state in s_t to states in s_{t-1}, we have the approximation

$$\left(1 \;-\; \frac{c_t}{N_{t-1}}\right)^{n_t} \;=\; \left(\left(1 \;+\; \frac{-c_t}{N_{t-1}}\right)^{N_{t-1}}\right)^{n_t/N_{t-1}} \;\approx\; \exp(-(c_t \, n_t)/N_{t-1}) \; . \tag{A248}$$

We obtain

$$n_{t-1} \;=\; (1 \;-\; \exp(-(c_t \, n_t)/N_{t-1})) \;\; N_{t-1} \; . \tag{A249}$$

For small n_t, we again have

$$n_{t-1} \;\approx\; c_t \, n_t \; . \tag{A250}$$

Therefore, for small $k - t$, we obtain

$$n_t \;\approx\; \prod_{\tau=t}^{k} c_\tau \;\approx\; \bar{c}^{k-t} \; . \tag{A251}$$

Thus, for small k the number a_k of states affected by $\boldsymbol{\omega}_k$ is

$$a_k \;=\; \sum_{t=0}^{k} n_t \;\approx\; \sum_{t=0}^{k} \bar{c}^{k-t} \;=\; \sum_{t=0}^{k} \bar{c}^{t} \;=\; \frac{\bar{c}^{k+1} - 1}{\bar{c} - 1} \;>\; \bar{c}^{k} \; . \tag{A252}$$

Consequently, for small k the number a_k of states affected by $\boldsymbol{\omega}_k$ grows exponentially with k by a factor of $\bar{c}$. For large k, at a certain time $t > \hat{t}$, n_t has grown such that $c_t n_t > N_{t-1}$, yielding $\exp(-(c_t n_t)/N_{t-1}) \approx 0$, and thus

$$n_t \;\approx\; N_t \; . \tag{A253}$$

Therefore

$$a_k \;-\; a_{\hat{t}-1} \;=\; \sum_{t=\hat{t}}^{k} n_t \;\approx\; \sum_{t=\hat{t}}^{k} N_t \;\approx\; (k - \hat{t} + 1) \; \bar{N} \; . \tag{A254}$$

Consequently, for large k the number a_k of states affected by $\boldsymbol{\omega}_k$ grows linearly with k by a factor of $\bar{N}$. □

Therefore, we aim for decreasing the on-site variance $\boldsymbol{\omega}_k$ for large k, in order to reduce the variance. In particular, we want to avoid delayed rewards and provide the reward as soon as possible in each episode. Our goal is to give the reward as early as possible in each episode to reduce the variance of action-values that are affected by late rewards and their associated immediate and local variances.